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Effect of exercise therapy on neuromuscular activity and knee strength in female adolescents with patellofemoral pain—An ancillary analysis of a cluster randomized trial

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ABSTRACT

Background: Female adolescents with patellofemoral pain are characterized by altered neuromuscular knee control and reduced maximal quadriceps torque. The purpose of this study is to investigate whether exercise therapy and patient education are associated with larger improvements in neuromuscular knee control and maximal quadriceps torque compared with patient education alone.

Methods: This is an ancillary analysis of a cluster randomized controlled trial investigating the effect of patient education and exercise therapy on self-reported recovery in 121 adolescents with patellofemoral pain. A random subsample of 57 female adolescents was included and tested at baseline and after 3 months. Neuromuscular control of the knee was quantified as the complexity of surface electromyography of the vastus lateralis and vastus medialis during stair descent. Secondary outcomes were complexity of knee flexion/extension kinematics and maximal quadriceps torque.

Findings: There was an 8–15% greater decrease in the complexity of surface electromyography suggesting an improvement in neuromuscular knee control among those randomized to exercise therapy ($0.08 < p < 0.30$). Adolescents randomized to exercise therapy had a 0.28-Nm/kg (95% CI: 0.05–0.52; $p = 0.02$) larger increase in maximal quadriceps torque.

Interpretation: Female adolescents randomized to patient education and exercise therapy had a significantly larger increase in maximal quadriceps torque and greater improvement in neuromuscular knee control during stair descent than those receiving patient education alone. This suggest that exercise therapy has an effect not only on self-reported outcome measures but also on objective measures of thigh muscle function in female adolescents with patellofemoral pain.

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1. Introduction

Self-reported knee pain is highly prevalent among adolescents (Fairbank et al., 1984; Rathleff et al., 2013b; Stovitz et al., 2008). Cross-sectional studies show that 19–31% of adolescents report knee pain (Fairbank et al., 1984; Rathleff et al., 2013e; Vahasarja, 1995). One of the most common knee conditions among adolescents is patellofemoral pain (PFP), which has a population prevalence of 6–7% (Molgaard et al., 2011; Rathleff et al., 2013c). The most frequent self-reported symptoms are diffuse peri-patellar and retro-patellar localized pain (Haim et al., 2006). Pain is typically provoked by activities of daily living that require loading on a flexed knee such as stair descent or squatting (Haim et al., 2006).

Recently, we observed a higher complexity (i.e., higher sample entropy) of the muscle activity of the vastus lateralis (VL) and the knee kinematics during stair descent in female adolescents with PFP (Rathleff et al., 2013d). Higher sample entropy values express a higher complexity of the signal which is thought to be suggestive of altered neuromuscular control (Hausdorff et al., 1995; Lamoth et al., 2004). Self-reported pain intensity was positively associated ($r^2 = 0.11$, $p = 0.01$) with the complexity of surface electromyography (sEMG) of VL. This indicates that adolescents with PFP with the highest amount of pain displayed greater signal irregularity or unpredictability of sEMG compared to

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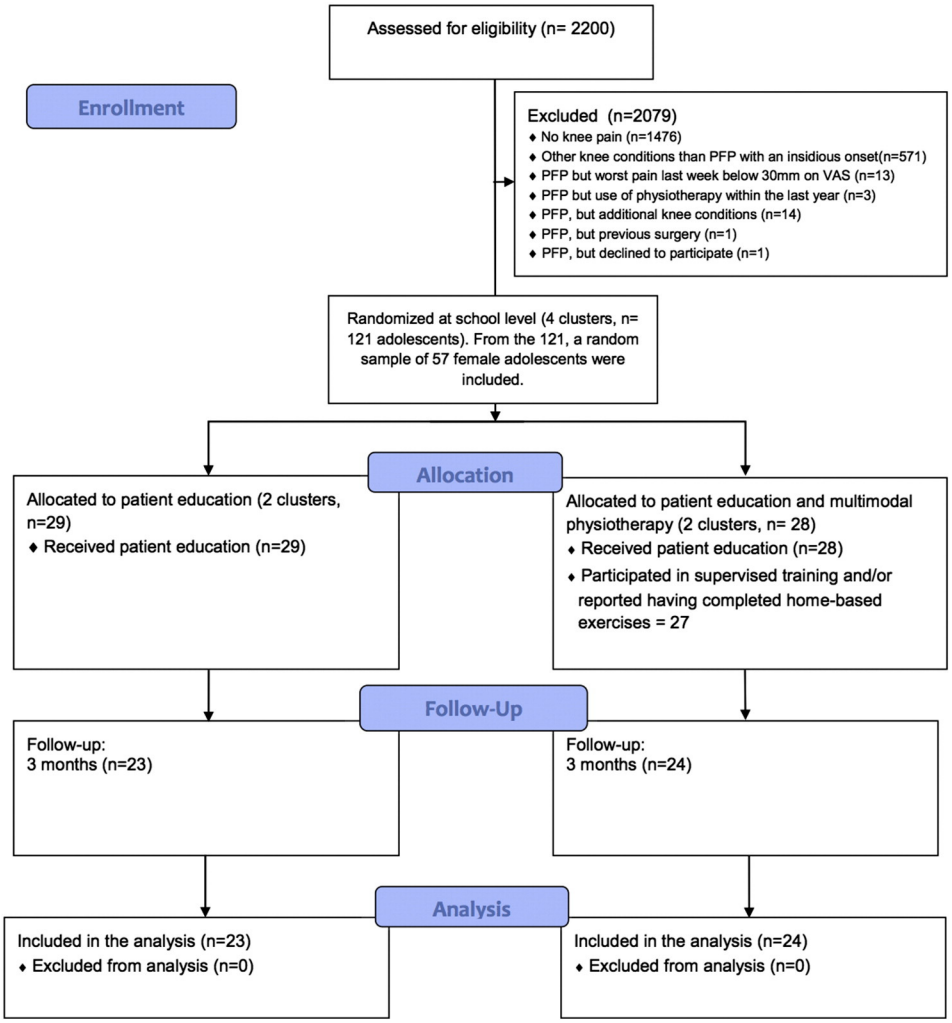

Fig. 1. Flow chart.

pain-free controls. These results suggest that rehabilitation is needed and should focus on both restoring neuromuscular control and increasing muscle strength (Rathleff et al., 2013d).

Two of the core elements in exercise therapy are neuromuscular training, challenging the control of the lower extremity, and strength training. One of the aims in the treatment of PFP is to target strength and neuromuscular control deficits that may help reduce patellofemoral loading. However, it is not known whether exercise therapy can improve neuromuscular knee control and increase quadriceps strength among adolescents with PFP. Therefore, the purpose of this ancillary analysis was to investigate whether 3 months of exercise therapy and patient education were associated with larger positive changes in neuromuscular control compared to patient education alone. We hypothesized that adolescents randomized to additional exercise therapy would show larger improvements in neuromuscular knee control after 3 months treatment compared to adolescents randomized to patient education alone.

2. Methods

This is an ancillary analysis of a cluster randomized controlled trial investigating the effect of exercise therapy with or without the addition of patient education on self-reported recovery (see description of recovery in the section "Self-reported recovery" (Rathleff et al., 2015). The main study showed that adolescents randomized to patient education and exercise therapy were more likely to be recovered at 3 to 24 month follow-up (Rathleff et al., 2014b). The adolescents included in this ancillary report were a randomly selected subsample of female adolescents from the original cluster randomized study (Rathleff et al., 2015). All adolescents were recruited from a population-based cohort (Adolescent Pain in Aalborg 2011, the APA2011-cohort) (Rathleff et al., 2012). The study was approved by the local ethics committee of North Denmark Region and all adolescents provided informed consent.

2.1. Recruitment

In the original study, 2846 students aged 15–19 years from four schools were invited to answer an online questionnaire concerning musculoskeletal pain and physical activity (Fig. 1) (Rathleff et al., 2012). The students who reported knee pain were contacted by telephone and offered a clinical examination by an experienced rheumatologist if they fulfilled the following criteria: pain for more than 6 weeks; pain felt anteriorly around the patella or diffusely around the knee; no treatment within the previous 12 months; and no previous knee surgery and a history of insidious onset of knee pain.

During the subsequent clinical examination, the students were diagnosed with PFP if they met the following criteria, which are in accordance with a previous clinical trial (Cowan et al., 2002):

- Insidious onset of anterior knee or retro-patellar pain for more than 6 weeks and provoked by at least two of the following situations:

prolonged sitting or kneeling, squatting, running, hopping, or stair walking
- Tenderness on palpation of the patella, or pain with stepping down or double leg squatting
- Worst pain during the previous week above 30 mm on a 100-mm visual analog scale (VAS)

Exclusion criteria were concomitant injury or pain from the hip, lumbar spine, or other knee structures such as the patella tendon, tibial tuberosity, or quadriceps tendon. Adolescents with previous knee surgery, patellofemoral instability, knee joint effusion, use of physiotherapy for treating knee pain within the previous year, or weekly use of anti-inflammatory drugs were also excluded.

From these, 153 were diagnosed with PFP and 122 were eligible to enter the original study, and 121 adolescents agreed to participate. The sample size for this ancillary analysis was based on our primary hypothesis and aimed at detecting a 15% larger improvement in SaEn from VL during stair descent among those randomized to patient education and exercise therapy (corresponding to a SaEn value of 0.008) compared to those randomized to patient education alone. Using a standard deviation of 0.01 and power of 80% and an alpha of 5%, at least 25 participants were needed in each group. We aimed at recruiting 60 to account for dropouts. Female adolescents with PFP were randomly selected for this sub-study using a random number generator and their project identification number. We contacted 77 female students with PFP to ensure the required sample of 60 female adolescents with PFP. Seventeen of them did not wish to participate, and three adolescents with PFP failed to attend on the day of testing. Therefore, we ended up with 57 female adolescents.

We only included females as the prevalence of PFP is higher among females than males (Molgaard et al., 2011). In addition, we assumed that including both sexes could be seen as a confounder since females are reported to have a lower complexity of movement (Rathleff et al., 2010; Svendsen et al., 2011) and may exhibit different movement patterns than males (Willy et al., 2012).

2.2. Outcomes

Outcome measurements were collected from the most painful knee in adolescents with PFP. A single rater, blinded to group allocation, conducted all measurements. The rater was a physiotherapist working as a research assistant and had approximately 1 year of experience in collecting sEMG and strength outcomes. Primary outcome was the neuromuscular knee control quantified as the complexity of sEMG of the VL and VM during stair descent while secondary outcome measures included complexity of knee flexion/extension kinematics and maximal quadriceps torque.

Bipolar surface electrodes (Ambu A/S, Neuroline, Ballerup, Denmark) were placed on VM and VL with an inter-electrode distance of 2 cm using previously described locations (Rathleff et al., 2013d). One foot-switch was placed under the heel and one under the hallux to identify heel-strike and toe-off (Rathleff et al., 2013d). The foot switches, sEMG signals, and knee kinematics were sampled synchronously at 2 kHz using a 12-bit A/D converter (Biovision, Werheim, Germany) and stored on disk. Before analysis, the sEMG signals were digitally band-pass filtered (10–400 Hz, fourth-order Butterworth) and down sampled to 1 kHz. A Notch filter (second-order Butterworth band stop with 1-Hz rejection width) centered at the power line frequency (50 Hz) was used when necessary to remove line interference.

After positioning the electrodes, the isometric strength of the quadriceps was measured as the maximal quadriceps torque at 60 degrees knee flexion (Fig. 2). Maximal quadriceps torque was measured using a portable handheld dynamometer (HHD) that was fixed to the examination table using a non-elastic strap (Rathleff et al., 2013a, 2013d). The HHD was a Power track II commander (Chiroform, Viborg, Denmark) manipulated by an experienced physiotherapist. The highest value of four consecutive measurements was used for the data analysis. A 1-min rest period was given between each trial. Data from the HHD were converted to torque values (Nm) by multiplying force (N) values from the HHD by the distance from the medial joint line at the knee to the position of the HHD. Torque values were divided by body weight to obtain the normalized quadriceps torque (Nm/kg). Muscle activity was measured during strength testing to allow for normalization of sEMG signals.

Fig. 2. Setup for testing maximal quadriceps torque.

After the maximal quadriceps torque measurements, an electronic uniaxial goniometer (BioVision, Wehrheim, Germany) was secured around the tibia and femur. Before analysis, the knee kinematics were digitally low-pass filtered (20 Hz, fourth-order Butterworth) and down sampled to 400 Hz.

2.3. Stair descent

The stairway consisted of two sets of 12 steps separated by a short landing. Subjects took approximately four steps on level ground before starting the stair descent. The participants were given one practice trial and were told to walk at their normal pace. Controlling the timing of stair descent would have changed the complexity of the sEMG and knee kinematics as shown earlier for gait in healthy subjects (Jordan et al., 2007). Thus, we did not control the speed of stair descent.

2.4. Randomization

Randomization was done after baseline testing. Randomization was performed at school (cluster) level, and the four schools were randomized to patient education with or without exercise therapy using a computer-generated sequence generated by the research team.

2.5. Interventions

2.5.1. Patient education

One physiotherapist delivered the patient education in the two clusters randomized to patient education alone. Patient education was standardized, lasted for about 30 min, and covered the following topics: reasons for pain; pain management; how to modify physical activity; how to return slowly to sports; how to cope with knee pain; information on optimal knee alignment during sit-to-stand, standing, walking, stair walking, and cycling; and responses to questions from the

adolescent or the parents. Adolescents also received this information in an eight-page leaflet.

2.5.2. Patient education and exercise therapy

The exercise therapy was based on previous trials (Crossley et al., 2002, 2008) and consisted of a combination of supervised group training sessions and unsupervised home-based exercises. The supervised intervention was offered three times a week on school premises immediately after the end of the school day for a period of 3 months.

The supervised training sessions consisted of neuromuscular training of the muscles around the foot, knee, and hip, strength training for the knee and hip, patellofemoral soft tissue mobilization, and stretching of the muscles around the hip and knee. To progressively match the exercise level to the performance level of each participant, all exercises were available in multiple levels of difficulty (Rathleff et al., 2012). All adolescents started with exercises at level one and progressed from there. The progression followed previously described rules (Ageberg et al., 2010; Rathleff et al., 2012). The unsupervised home-based exercises consisted of approximately 15 min of quadriceps and hip muscle retraining and stretching. The exercises were to be performed each day except on the days of supervised group training. Additional details on the interventions can be found in (Rathleff et al., 2012, 2015). Adherence to the supervised sessions was recorded as the number of times each adolescent participated. Adherence to home exercises was monitored by weekly follow-ups using Short Message Service (SMS).

2.6. Self-reported recovery

Recovery was measured on a seven-point Likert scale ranging from "completely recovered" to "worse than ever" in line with a similar randomized trial among this patient population (van Linschoten et al., 2009). The adolescents were asked to rate their knee pain now compared to before they received treatment. Patients were categorized as recovered if they rated themselves as "fully recovered" or "strongly recovered," whereas those who rated themselves as "slightly recovered" to "worse than ever" were categorized as not recovered.

2.7. Data analysis

The data analysis follows our previous cross-sectional study that used the same types of outcomes (Rathleff et al., 2013d). Only the stance phase of stair descent was analyzed as patients with PFP primarily report pain during the stance phase of stair descent. The stance phase was defined as toe-on until toe-off. The time segment on the intermediate platform between the first 12 steps and the next 12 steps as well as the first and last step during descent were excluded from the data analysis by using a manual trigger during data collection. Toe-on and toe-off were not always detectable, resulting in the number of stance phases varying from 9 to 11. To account for changes in sEMG during stair descent, the total number of stance phases was divided into three intervals representing the start, middle, and end, corresponding to 0%, 50%, and 100% of stair descent, respectively. The maximum voluntary electrical activation (MVE) was calculated as the highest sEMG root mean square (RMS) calculated over 250-ms epochs moving in steps of 100 ms. Lastly, the normalized values were averaged over all recorded strides for each time segment (nRMS). Likewise, the kinematics data was divided into three intervals representing start–middle–end of stair descent.

Sample entropy was used to quantify the complexity of goniometer and sEMG time series during stair descent. The stance phases were divided into three intervals representing the start–middle–end corresponding to 0%–50%–100% of stair descent. These 3 intervals were composed of 4 concatenated stance phases of an average of 776 (SD 95) and 1552 (SD 190) samples for goniometer and sEMG data, respectively. Entropy assesses the probability that equal sequences of length m remain similar after a time increment. The degree of similarity is fixed by the tolerance r of the time series. In the present study, m and r were set to 2% and 10%, respectively, of the SD of the time series in agreement with theoretical and experimental studies (Pincus et al., 1991; Richman and Moorman, 2000; Sondergaard et al., 2010). Sample entropy is a unitless non-negative number. Lower SaEn values indicate a less irregular, unpredictable, and complex structure of the time series. On the contrary, higher SaEn values correspond to more irregular, unpredictable, and complex signal (Richman and Moorman, 2000; Svendsen et al., 2011). Our previous study showed that female adolescents with PFP were characterized by significantly higher SaEn (higher complexity) from VL and knee flexion/extension kinematics during stair descent compared to pain-free adolescents. Changes in nRMs and SaEn were used to depict altered neuromuscular control of the knee joint. In the context of the current study, we interpret decreases in amplitude (nRMS) and complexity (SaEn) as improvement in neuromuscular knee control (Rathleff et al., 2013d).

2.8. Statistics

Between-group comparison of sample entropy from VL, VM, knee kinematics, and maximal quadriceps torque was performed using linear regression for repeated measurement with robust standard errors adjusted for within-cluster correlations for schools to account for the cluster randomized design (Williams, 2000). Associations between changes in neuromuscular knee control, changes in pain, and adherence to exercises were analyzed using the same approach. The t-values from the linear regression were used to calculate the effect size (Cohen's d). A $p < 0.05$ was considered significant. All calculations were conducted using Stata version 11.0.

3. Results

3.1. Participant characteristics

At baseline, 57 female adolescents were included (Tables 1 and 2). At 3 months follow-up 47 adolescents were available for testing, corresponding to a follow-up rate of 82%. The adolescents randomized to patient education and exercise therapy reported a mean of 43 training sessions (range 0–87) during the 3 months.

3.2. Dropout analysis

The 11 subjects (six from patient education and five from patient education and exercise therapy) that did not attend the 3-month follow-up appointment, showed no significant or relevant difference in baseline SaEn (SaEn from VM or VL < 0.001, $p > 0.15$), maximal quadriceps torque (0.15 [Nm/kg], $p = 0.32$), self-reported outcomes (KOOS score < 6, $p > 0.26$), or demographics (age < 0.1 year, weight < 0.1 kg, height < 4 cm, BMI < 0.1, $p > 0.73$) compared to the 47 who completed the 3-months follow-up.

3.3. Surface EMG

There was no significant difference between groups in the change in sample entropy from sEMG of VL and VM during stair descent. However, consistently across all three intervals, subjects randomized to patient education and exercise therapy showed larger decreases in sample entropy from VL (interval 1 = −0.007, $p = 0.25$, $d = 0.41$) (interval 2 = −0.008, $p = 0.009$, $d = 0.72$) (interval 3 = −0.004, $p = 0.30$, $d = 0.36$) and from VM (interval 1 = −0.007, $p = 0.08$, $d = 0.75$) (interval 2 = −0.007, $p = 0.11$, $d = 0.67$) (interval 3 = −0.005, $p = 0.29$, $d = 0.38$) (Fig. 3). This corresponds to an 8–15% larger decrease (effect size of 0.38–0.75) for both VL and VM from baseline to 3 months follow-up among subjects randomized to patient education and exercise therapy compared to those receiving patient education alone.

Adolescents randomized to patient education and exercise therapy showed a larger decrease in nRMS from VL (interval 1 = 2.6%, $p =$

Table 1
Demographics at baseline.

	Patient education, $n = 29$	Patient education and exercise therapy, $n = 28$	All
Age [years]	17 (16–18)[a]	17 (16–18)*	17 (16–18)[a]
Height [cm]	168.1 (4.8)	169.3 (5.8)	165.7 (5.4)
Body mass [kg]	59.4 (5.7)	58.0 (6.4)	58.7 (6.0)
BMI [kg/m^2]	21.0 (2.0)	20.2 (1.7)	20.6 (1.9)
Pain duration [months]	36 (22.5–57)[a]	24 (17–48)[a]	36 (21–51)[a]
VAS worst pain	50.5 (43.2–57.8)	52.2 (44.2–60.2)	51.3 (46.0–56.7)

Demographics presented as mean values (95% CI).
[a] Median and interquartile range.

0.14, $d = 0.59$) (interval 2 = 1.3%, $p = 0.49$, $d = 0.23$) (interval 3 = 3.0%, $p = 0.04$, $d = 1.03$.) but not from VM (interval 1 = 5.2%, $p = 0.36$, $d = 0.32$) (interval 2 = 4.3%, $p = 0.38$ $d = 0.31$) (interval 3 = 4.7%, $p = 0.33$, $d = 0.35$) compared to those receiving patient education alone.

3.4. Quadriceps torque

At 3 months follow-up, the maximal quadriceps torque was 2.17 Nm/kg (1.96–2.39) and 2.54 Nm/kg (2.27–2.82), in the groups without and with exercise therapy, respectively. The group receiving patient education and exercise therapy had a 0.28-Nm/kg (95% CI: 0.05–0.52, $p = 0.02$) larger increase in quadriceps torque compared to the group receiving patient education alone. This corresponds to an effect size of 0.71.

3.5. Range of motion

The change in sample entropy from knee flexion/extension kinematics from baseline to 3 months follow-up did not differ significantly between groups (interval 1 = −0.006, $p = 0.47$, $d = 0.21$) (interval 2 = −0.003, $p = 0.72$, $d = 0.11$) (interval 3 = −0.005, $p = 0.53$, $d = 0.19$) (Fig. 4). Likewise, there was no change in the knee range of motion between groups (interval 1 = −1.7, $p = 0.40$, $d = 0.29$) (interval 2 = −1.2, $p = 0.45$, $d = 0.25$) (interval 3 = 0.0, $p = 1.00$, $d = 0.00$).

3.6. Association between pain, strength, adherence, and sample entropy

There was no association between the change in SaEn from VL or VM and the change in pain from baseline to follow-up ($r^2 = 0.02$ and 0.03, $p > 0.67$). Likewise, there was no significant difference between adolescents who reported themselves "recovered" and adolescents who had "not recovered" ($p > 0.81$). There was no association between change in maximal quadriceps torque and change in SaEn from VL or VM ($r^2 = 0.04$ and 0.06, $p > 0.12$).

Table 2
Outcome measures at baseline.

	Patient education, $n = 28$, mean (95% CI)	Patient education and exercise therapy, $n = 29$, mean (95% CI)
Sample entropy vastus lateralis[a]	0.054 (0.051–0.057)	0.056 (0.053–0.059)
Sample entropy vastus medialis[a]	0.046 (0.038–0.054)	0.054 (0.052–0.056)
Sample entropy knee kinematics	0.0059 (0.0057–0.0062)	0.0064 (0.0047–0.0081)
Range of motion for knee kinematics[a]	67.6 (64.8–70.4)	66.5 (54.7–78.4)
nRMS[b] from vastus lateralis[a]	0.29 (0.26–0.32)	0.27 (0.21–0.34)
nRMS[b] from vastus medialis[a]	0.35 (0.28–0.42)	0.31 (0.21–0.41)
Normalized maximal isometric quadriceps torque [Nm/kg]. Max of four trials	2.26 (2.09–2.43)	2.27 (2.08–2.47)

[a] Presented as mean across the three cycles and adjusted for cluster.
[b] Normalized root mean square.

An explorative within-group analysis among adolescents randomized to patient education and exercise therapy showed that the total number of training sessions was significantly associated with the decrease in sample entropy from sEMG from VL ($r^2 = 0.16$, $p = 0.045$) but not from VM ($r^2 = 0.05$, $p = 0.44$).

4. Discussion

This is the first study to investigate the effect of exercise therapy on neuromuscular knee control and maximal quadriceps torque in female adolescents with PFP. We found that exercise therapy were associated with larger increases in maximal quadriceps torque compared to education alone, but had no significant effect on neuromuscular control during stair descent. However, across all intervals and across both VL and VM there were tendencies towards larger decreases in the complexity and amplitude of sEMG suggesting an improvement in neuromuscular control. These findings indicate that exercise therapy has an effect on both self-reported outcome measures as previously shown, and on objective measures of thigh muscle function among adolescents suffering from long-standing PFP.

4.1. Exercise therapy for adolescents with PFP

The active exercises used in the current study were a combination of strength training at 10–12 RM and exercises that challenged control of the foot, knee, and hip. Therefore, we expected a positive change in neuromuscular knee control depicted by changes in complexity and amplitude of sEMG during stair descent. The exercises were progressive in nature, meaning that both the load and the difficulty of the exercises were progressed throughout the trial depending on how the adolescents responded to the exercises. As shown in our previous publication on self-reported outcome measures, compliance was an issue, and none of the adolescents showed good compliance with the supervised training sessions. This may be one of the reasons for the small difference between groups. This is supported by the significant association between adherence and change in our primary outcome (SaEn from VL). However, it is not possible to infer clinical relevance of our findings on the primary outcome. In a recent study, Samani et al. (2015) have reported absolute and relative reliability of SaEn computed on kinematic recordings (Samani et al., 2015). The minimal detectable change in SaEn reported by Samani et al. (2015) is larger than the changes in SaEn we report here after exercise therapy. Thus, it is not possible to infer clinical relevance of our findings on this outcome.

In our previous cross-sectional study, we observed an association between self-reported pain and neuromuscular control, which led us to speculate that improvements in knee pain would be associated with improvement in neuromuscular control (Rathleff et al., 2013d). However, we found no association between improvements in pain and positive changes in neuromuscular control. This could mean that even though pain is associated with neuromuscular control cross-sectionally, self-report, and objectively measured neuromuscular control may improve through different pathways and may not be causally linked. The present randomized controlled study aimed at investigating these aspects in details.

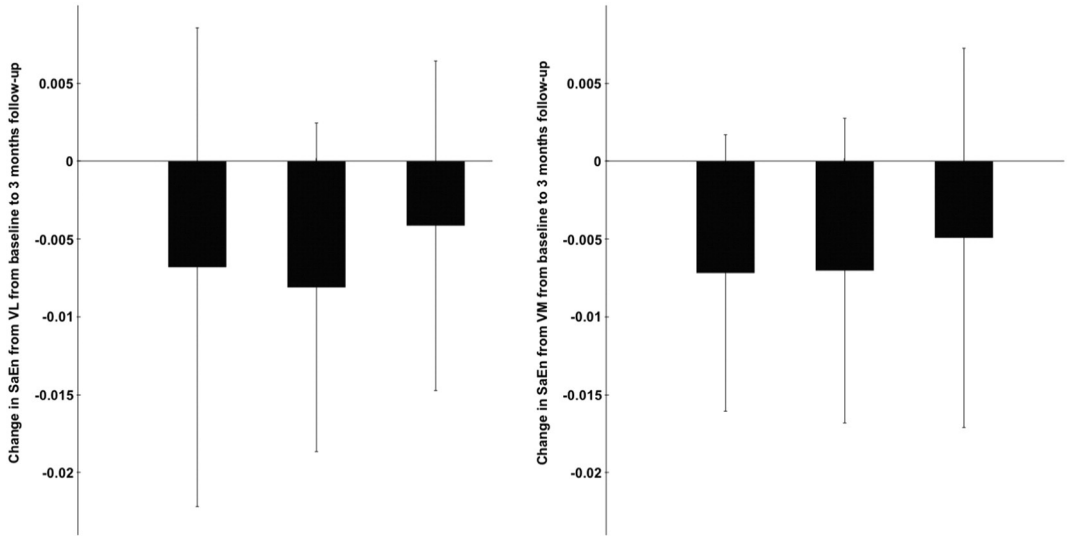

Fig. 3. Change in sample entropy (SaEn) from vastus lateralis (VL) and vastus medialis (VM) from baseline to follow-up between the two groups presents three intervals representing the start–middle–end corresponding to 0%–50%–100% of stair descent. Negative values indicate that the group randomized to patient education and exercise therapy had a larger decrease in SaEn compared to the group randomized to patient education alone. Effect sizes: VL (interval 1 = −0.007, $p = 0.25$, $d = 0.41$) (interval 2 = −0.008, $p = 0.009$, $d = 0.72$) (interval 3 = −0.004, $p = 0.30$, $d = 0.36$) and from VM (interval 1 = −0.007, $p = 0.08$, $d = 0.75$) (interval 2 = −0.007, $p = 0.11$, $d = 0.67$) (interval 3 = −0.005, $p = 0.29$, $d = 0.38$). Error bars show the adjusted 95% CI confidence intervals.

4.2. Neuromuscular control, strength, and knee kinematics during stair descent

No previous studies have investigated exercise-induced changes in neuromuscular control and strength among adolescents with PFP. However, previous studies have investigated different measures of neuromuscular control and knee kinematics among adults with PFP. Cowan et al. showed that multimodal intervention, similar to that used in the current study, was associated with improvements in muscle onset of the VM compared to VL during a functional stair stepping task compared to a placebo treatment (Cowan et al., 2002). Even though muscle onset and SaEn require different processing of sEMG both metrics are thought to reflect aspects of neuromuscular control.

Two studies have previously investigated changes in knee kinematics during running after a hip strengthening program and found no change in knee kinematics, despite significant symptom reduction (Earl and Hoch, 2011; Ferber et al., 2011). Interestingly, Ferber et al. found a larger stride-to-stride knee joint variability when comparing patients with PFP to pain-free controls. After a 3-week rehabilitation they observed a significant reduction in stride-to-stride knee joint variability which made the individuals with PFP more similar to healthy pain-free controls (Ferber et al., 2011). We did not observe significant

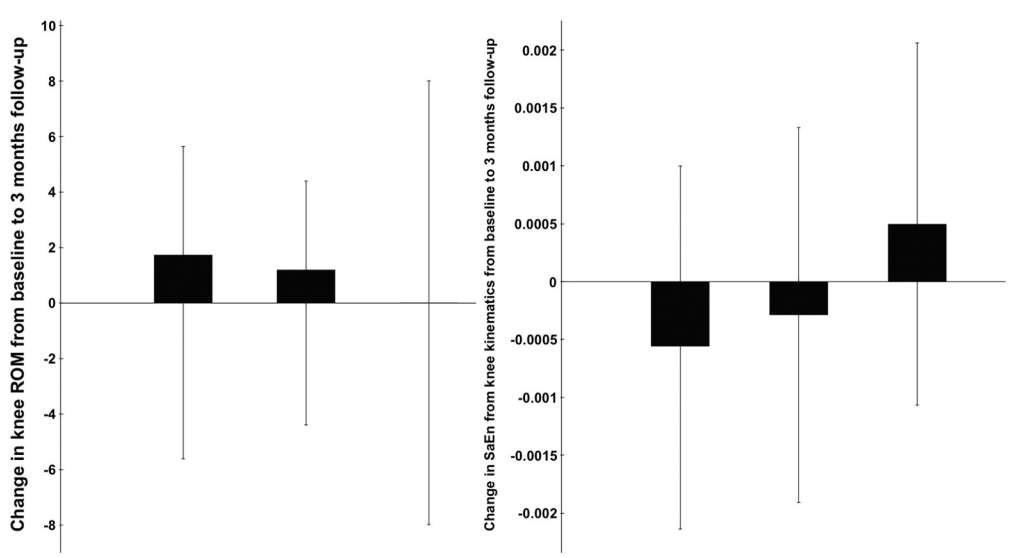

Fig. 4. Change in sample entropy (SaEn) from knee kinematics and knee range of motion (ROM) from baseline to follow-up between the two groups. Negative values indicate that the group randomized to patient education and exercise therapy showed a larger decrease in SaEn or ROM compared to those randomized to patient education alone. Knee range of motion (interval 1 = −1.7, $p = 0.40$, $d = 0.29$) (interval 2 = −1.2, $p = 0.45$, $d = 0.25$) (interval 3 = 0.0, $p = 1.00$, $d = 0.00$). Sample entropy from knee range of motion: (interval 1 = −0.006, $p = 0.47$, $d = 0.21$) (interval 2 = −0.003, $p = 0.72$, $d = 0.11$) (interval 3 = −0.005, $p = 0.53$, $d = 0.19$). Error bars show the adjusted 95% CI confidence intervals.

changes in complexity of the knee kinematics but non-significant changes in complexity of sEMG from VM and VL. Collectively, this may indicate that neuromuscular knee control may be improved by exercise, but adequately powered trials are needed to confirm this.

In our previous cross-sectional study (Rathleff et al., 2013d), we showed that female adolescents with PFP had a lower maximal quadriceps torque compared to adolescents without knee pain (2.27 vs. 2.80 Nm/kg). After the intervention period, adolescents allocated to exercise therapy and patient education increased their strength to 2.54 Nm/kg. This shows for the first time that maximal quadriceps torque can be increased among adolescents with PFP within a short period of time. These novel findings are similar to Clark et al., who found that exercise and education were associated with larger, but insignificant, increases in quadriceps strength compared to patella taping among adults with PFP (Clark et al., 2000). Furthermore, the relative increase in maximal quadriceps torque seen in our study was similar to those of previous trials in adults with PFP, at around 6–20% (Kaya et al., 2010; Witvrouw et al., 2000). This is important as increasing quadriceps strength may be associated with a better long-term outcome (Natri et al., 1998).

4.3. Limitations and strengths

Our a priori sample size calculation was based on detecting a 15% difference between groups in the complexity of sEMG from VL. As expected, the results showed an 8–15% larger improvement in neuromuscular control of VM and VL among adolescents randomized to exercise therapy. However, the variance between subjects was larger than anticipated in our sample size calculation, which suggests that we were underpowered in the comparison of complexity of sEMG. Furthermore, the exercise dose (due to poor compliance with supervised exercise) was lower than anticipated. To further elucidate the changes in neuromuscular knee control and knee kinematics, future trials should consider ways to optimize compliance to increase the potential effect of exercise on neuromuscular knee control. Furthermore, it may be needed to control for the amount of contact with a therapist to reduce performance bias because of unequal therapist contact and further validate the use of SaEn as a proxy of neuromuscular control. A strength of the current study was the use of a blinded assessor. As highlighted in two recent reviews of strength measurements in individuals with PFP, most studies do not use blinded assessors (Lankhorst et al., 2013; Rathleff et al., 2014a). The assessor who positioned the electrodes and knee goniometer and collected maximal quadriceps torque was blinded as well as the person responsible for all data analysis. This reduced the risk of detection bias.

5. Conclusion

This ancillary analysis of a cluster randomized controlled trial showed a significantly larger increase in maximal quadriceps torque and a non-significant greater improvement in neuromuscular knee control during stair descent in female adolescents following patient education and exercise therapy compared with adolescents receiving patient education alone. These findings are important as they show that among adolescents suffering from long-standing PFP, exercise therapy may have an effect on objective measures of thigh muscle function as well as self-reported outcomes.

Conflicts of interest and source of funding

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